

DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Question 1.

Code:

Q1: Distribution Diagnosis

```
library(ggplot2)
```

```
response_time <- c(  
  118, 121, 125, 127, 129, 131, 134, 136, 138, 141,  
  144, 147, 151, 156, 162, 171, 184, 205, 249, 318  
)
```

```
df1 <- data.frame(response_time)
```

ECDF Plot

```
ggplot(df1, aes(x = response_time)) +  
  stat_ecdf(geom = "step") +  
  labs(  
    title = "ECDF of Response Time",  
    x = "Response Time (ms)",  
    y = "Cumulative Probability"  
  ) +  
  theme_minimal()
```

Normal Q-Q Plot

```
ggplot(df1, aes(sample = response_time)) +
```

```

stat_qq() +
stat_qq_line() +
labs(
  title = "Normal Q-Q Plot of Response Time",
  x = "Theoretical Quantiles",
  y = "Sample Quantiles"
) +
theme_minimal()

```

Skewness

```

mean(response_time)
median(response_time)

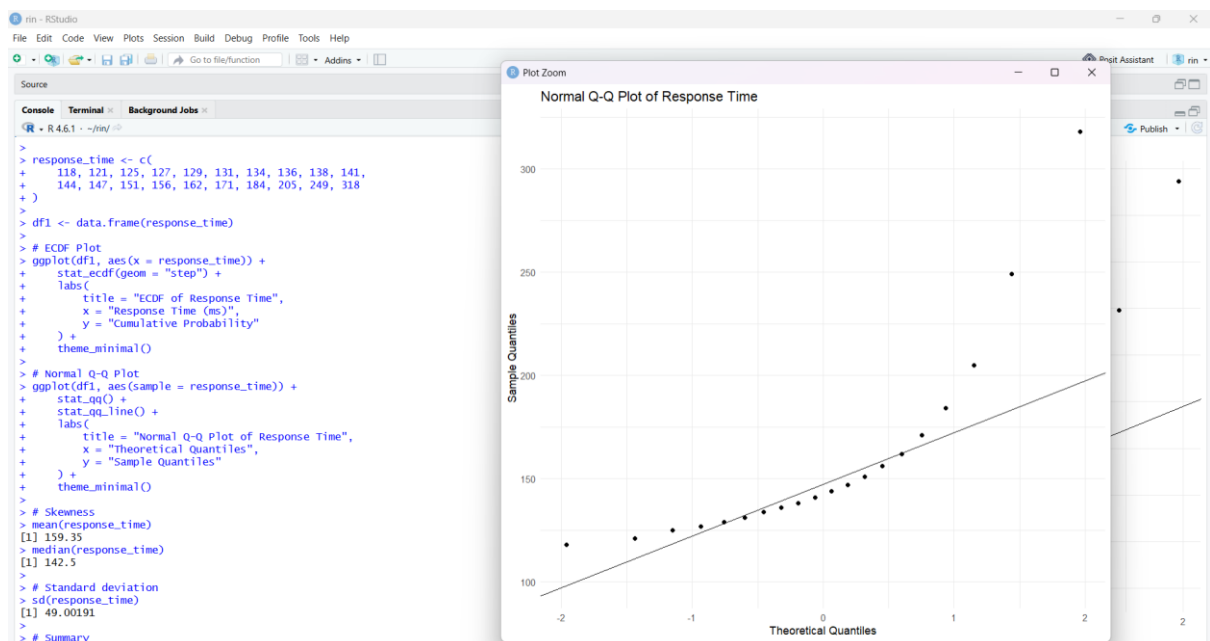
```

Standard deviation

```
sd(response_time)
```

Summary

```
summary(response_time)
```



Question 2:

Code:

Q2: Production Line Cycle Times

```
library(ggplot2)
```

```
library(ggribes)
```

```
L1 <- c(42, 44, 45, 46, 47, 48, 49, 50, 51, 53)
```

```
L2 <- c(38, 41, 44, 47, 50, 53, 56, 59, 62, 65)
```

```
L3 <- c(45, 46, 46, 47, 47, 48, 48, 49, 49, 50)
```

```
L4 <- c(40, 41, 42, 43, 44, 45, 48, 55, 61, 72)
```

```
df2 <- data.frame(
```

```
  Line = rep(c("L1", "L2", "L3", "L4"), each = 10),
```

```
  CycleTime = c(L1, L2, L3, L4)
```

```
)
```

```
# Boxplot comparison
```

```
ggplot(df2, aes(x = Line, y = CycleTime)) +
```

```
  geom_boxplot() +
```

```
  labs(
```

```
    title = "Cycle Time Comparison",
```

```
    x = "Production Line",
```

```
    y = "Cycle Time (minutes)"
```

```
  ) +
```

```
  theme_minimal()
```

```
# Violin / Bean-like distribution plot
```

```
ggplot(df2, aes(x = Line, y = CycleTime)) +
```

```
  geom_violin(trim = FALSE, fill = "lightblue") +
```

```

geom_jitter(width = 0.08, size = 2) +
labs(
  title = "Bean/Violin Plot of Cycle Times",
  x = "Production Line",
  y = "Cycle Time (minutes)"
) +
theme_minimal()

```

Vertical distribution using ridgeline

```

ggplot(df2, aes(x = CycleTime, y = Line, fill = Line)) +
  geom_density_ridges(alpha = 0.6) +
  labs(
    title = "Distribution of Cycle Times",
    x = "Cycle Time (minutes)",
    y = "Production Line"
  ) +
  theme_ridges()

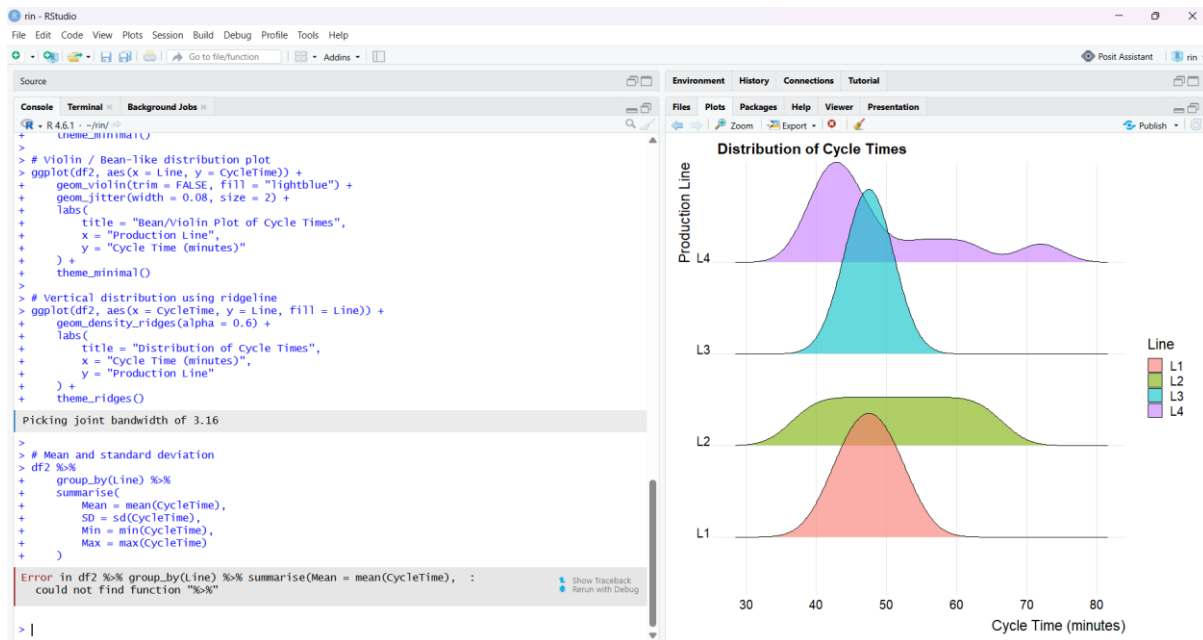
```

Mean and standard deviation

```

df2 %>%
  group_by(Line) %>%
  summarise(
    Mean = mean(CycleTime),
    SD = sd(CycleTime),
    Min = min(CycleTime),
    Max = max(CycleTime)
  )

```



Question 3:

Code:

Q3: Learning Mode Proportions

```
library(ggplot2)
```

```
library(tidyr)
```

```
df3 <- data.frame(
```

```
  Department = c("CSE", "ECE", "IT"),
```

```
  Online = c(52, 45, 30),
```

```
  Blended = c(28, 35, 40),
```

```
  Face_to_Face = c(20, 20, 30)
```

```
)
```

```
long3 <- pivot_longer(
```

```
  df3,
```

```
  cols = c(Online, Blended, Face_to_Face),
```

```
  names_to = "Learning_Mode",
```

```
  values_to = "Percentage"
```

```
)
```

```
# Side-by-side bar chart
```

```
ggplot(long3,  
  aes(x = Department, y = Percentage,  
    fill = Learning_Mode)) +  
geom_col(position = "dodge") +  
labs(  
  title = "Learning Mode by Department",  
  x = "Department",  
  y = "Percentage (%)"  
) +  
theme_minimal()
```

```
# 100% Stacked Bar Chart
```

```
ggplot(long3,  
  aes(x = Department, y = Percentage,  
    fill = Learning_Mode)) +  
geom_col(position = "fill") +  
scale_y_continuous(labels = scales::percent) +  
labs(  
  title = "100% Stacked Learning Mode Distribution",  
  x = "Department",  
  y = "Proportion"  
) +  
theme_minimal()
```

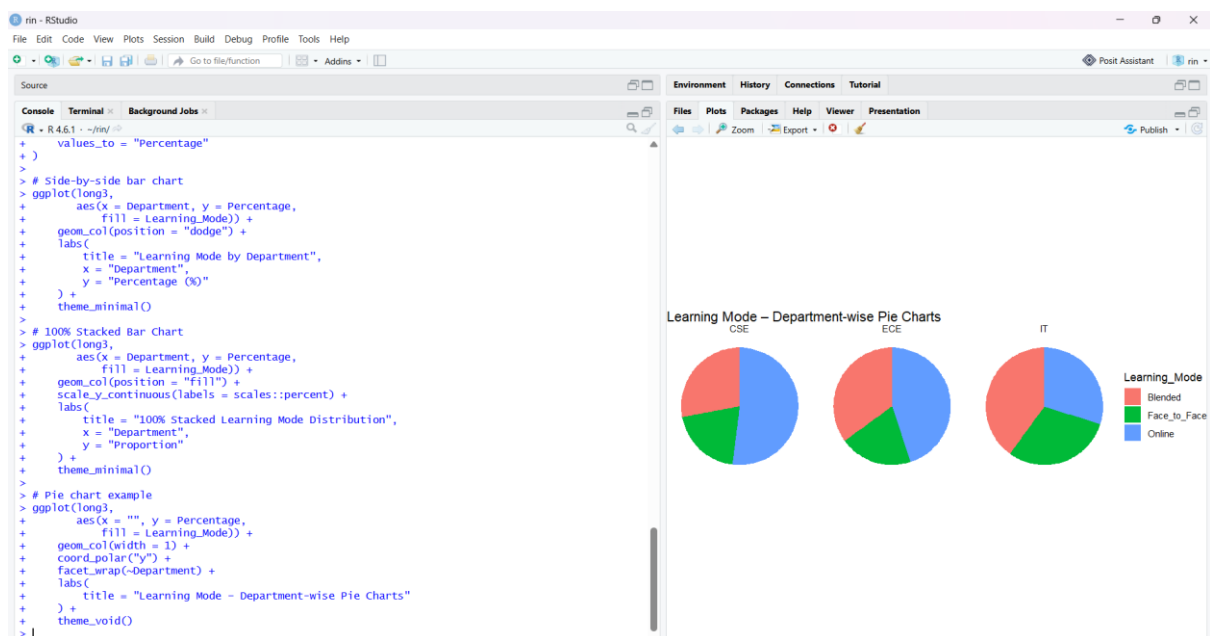
```
# Pie chart example
```

```
ggplot(long3,
```

```

aes(x = "", y = Percentage,
    fill = Learning_Mode)) +
geom_col(width = 1) +
coord_polar("y") +
facet_wrap(~Department) +
labs(
  title = "Learning Mode – Department-wise Pie Charts"
) +
theme_void()

```



Question 4:

Code:

Q4: E-commerce Revenue

```
library(ggplot2)
```

```
library(plotly)
```

```
library(treemap)
```

```
df4 <- data.frame(
```

```
  Category = c(
```

```

    "Electronics", "Electronics", "Electronics",
    "Home", "Home", "Home",
    "Fashion", "Fashion", "Fashion"
  ),
  Subcategory = c(
    "Mobiles", "Laptops", "Accessories",
    "Furniture", "Appliances", "Decor",
    "Men", "Women", "Kids"
  ),
  Revenue = c(24, 18, 8, 14, 10, 6, 7, 8, 5)
)

```

```

# -----

```

```

# Treemap

```

```

# -----

```

```

treemap(
  df4,
  index = c("Category", "Subcategory"),
  vSize = "Revenue",
  type = "index",
  title = "E-commerce Revenue Treemap",
  fontsize.labels = c(15, 10)
)

```

```

# -----

```

```

# Category totals

```

```

# -----

```



```

category_total <- aggregate(
  Revenue ~ Category,
  data = df4,
  sum
)

print(category_total)

# Percentage of total
category_total$Percentage <-
  category_total$Revenue /
  sum(category_total$Revenue) * 100

```

```

print(category_total)

```

```

# -----
# Sunburst
# -----

```

```

fig <- plot_ly(
  labels = c(
    "Total",
    "Electronics", "Mobiles", "Laptops", "Accessories",
    "Home", "Furniture", "Appliances", "Decor",
    "Fashion", "Men", "Women", "Kids"
  ),

```

```

parents = c(
  "",

```

```

    "Total", "Electronics", "Electronics", "Electronics",
    "Total", "Home", "Home", "Home",
    "Total", "Fashion", "Fashion", "Fashion"
  ),

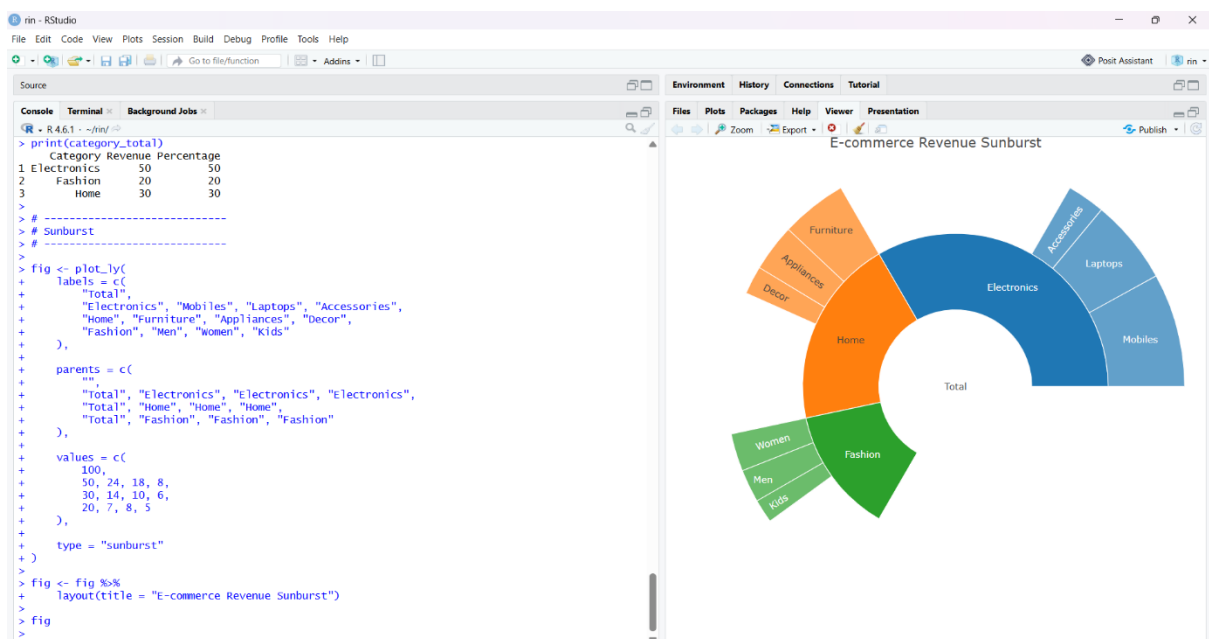
  values = c(
    100,
    50, 24, 18, 8,
    30, 14, 10, 6,
    20, 7, 8, 5
  ),

  type = "sunburst"
)

fig <- fig %>%
  layout(title = "E-commerce Revenue Sunburst")

```

fig



Question 5:

Code:

Q5: Smart City Mobility

```
library(dplyr)
```

```
library(networkD3)
```

```
library(ggplot2)
```

```
library(ggalluvial)
```

```
df5 <- data.frame(
```

```
  Origin = c(
```

```
    "Residential", "Residential",
```

```
    "Residential", "Residential",
```

```
    "Suburban", "Suburban",
```

```
    "Suburban", "Suburban",
```

```
    "Rural", "Rural",
```

```
    "Rural", "Rural"
```

```
),
```

```
  Mode = c(
```

```
    "Bus", "Bus",
```

```
    "Train", "Train",
```

```
    "Bus", "Bus",
```

```
    "Train", "Train",
```

```
    "Bus", "Bus",
```

```
    "Train", "Train"
```

```
),
```

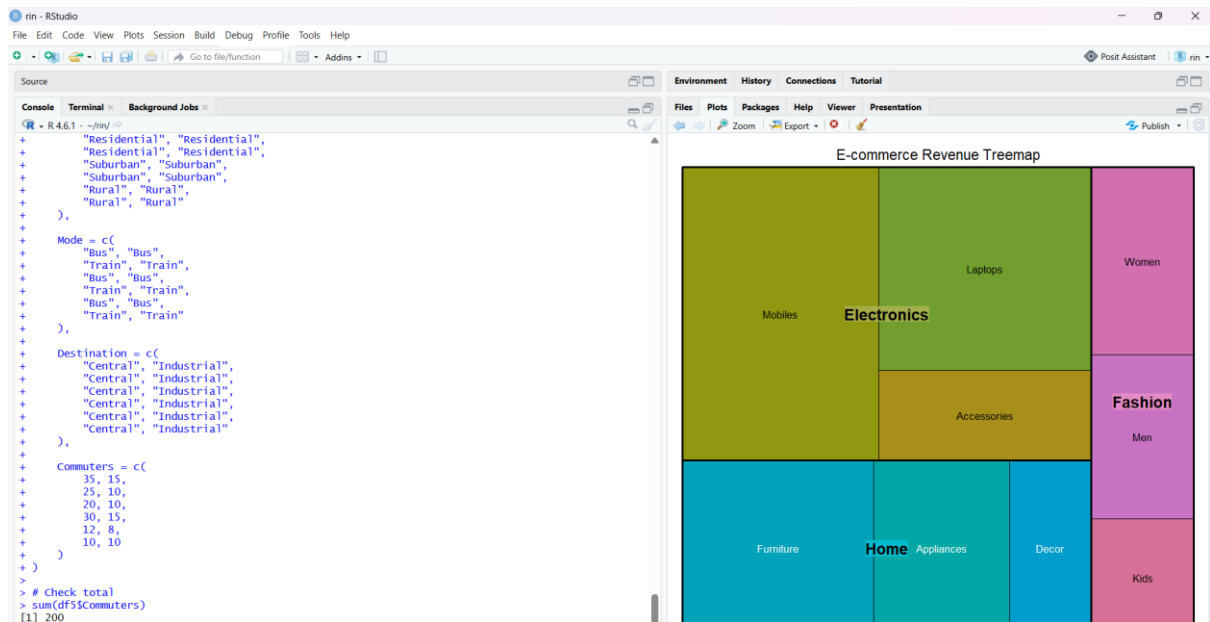
```
  Destination = c(
```

```
"Central", "Industrial",  
"Central", "Industrial",  
"Central", "Industrial",  
"Central", "Industrial",  
"Central", "Industrial",  
"Central", "Industrial"  
)
```

```
Commuters = c(  
  35, 15,  
  25, 10,  
  20, 10,  
  30, 15,  
  12, 8,  
  10, 10  
)  
)
```

```
# Check total
```

```
sum(df5$Commuters)
```



Sankey diagram:

Create nodes

```
nodes <- data.frame(
  name = unique(c(
    df5$Origin,
    df5$Mode,
    df5$Destination
  ))
)
```

Origin -> Mode

```
links1 <- df5 %>%
  group_by(Origin, Mode) %>%
  summarise(Value = sum(Commuters), .groups = "drop") %>%
  mutate(
    source = match(Origin, nodes$name) - 1,
    target = match(Mode, nodes$name) - 1
  ) %>%
  select(source, target, Value)
```

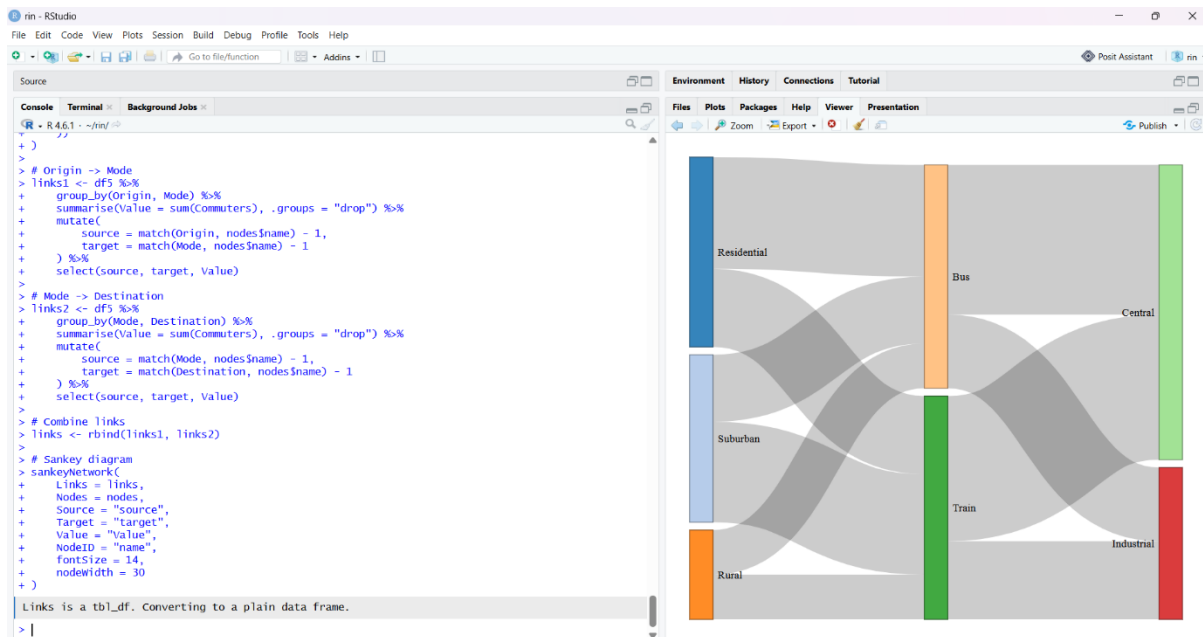
```

# Mode -> Destination
links2 <- df5 %>%
  group_by(Mode, Destination) %>%
  summarise(Value = sum(Commuters), .groups = "drop") %>%
  mutate(
    source = match(Mode, nodes$name) - 1,
    target = match(Destination, nodes$name) - 1
  ) %>%
  select(source, target, Value)

# Combine links
links <- rbind(links1, links2)

# Sankey diagram
sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "Value",
  NodeID = "name",
  fontSize = 14,
  nodeWidth = 30
)

```



Parallel Sets:

```

ggplot(
  df5,
  aes(
    axis1 = Origin,
    axis2 = Mode,
    axis3 = Destination,
    y = Commuters
  )
) +
  geom_alluvium(aes(fill = Mode), width = 1/12) +
  geom_stratum(width = 1/12, fill = "grey80", color = "black") +
  geom_text(
    stat = "stratum",
    aes(label = after_stat(stratum))
  ) +
  scale_x_discrete(
    limits = c("Origin", "Mode", "Destination"),
    expand = c(.05, .05)
  )

```

) +

labs(

title = "Parallel Sets – Commuter Flow",

x = "Category",

y = "Number of Commuters"

) +

theme_minimal()

